CM3121 – Advanced Organic Chemistry

Dr. Zhu Ye

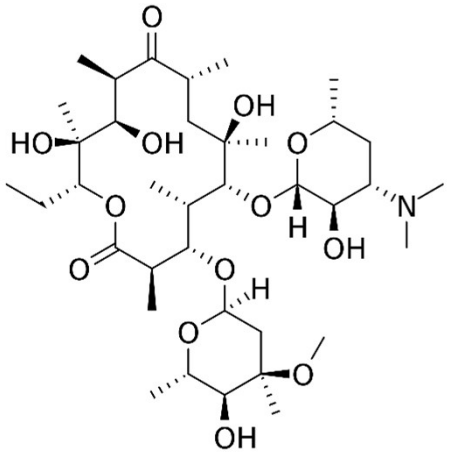
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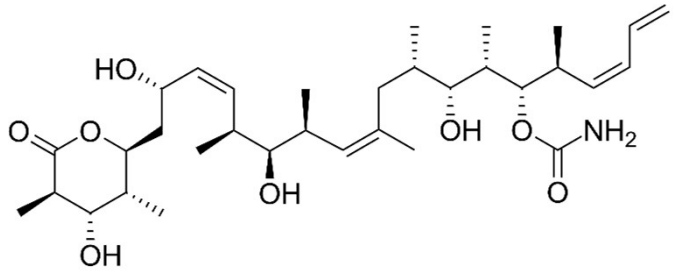
Aldol reactions

Readings: Solomons 13th ed, Chapter 19 and Clayden 2nd ed; Chapter 26

Natural products



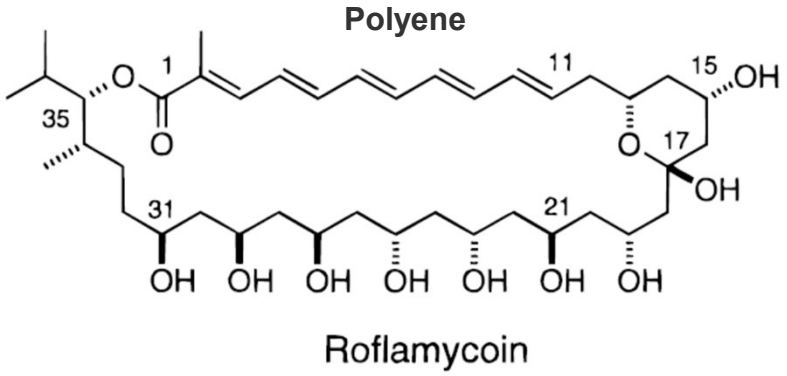
Erythromycin is an antibiotic used for the treatment of a number of bacterial infection



(+)-Discodermolide was found to be a potent inhibitor of tumor cell growth in several MDR cancer cell lines

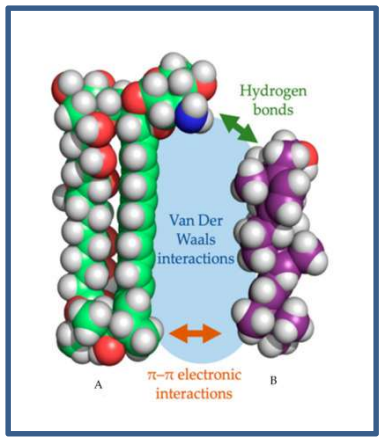
1. By Ed (Edgar181) - Own work, Public Domain, https://commons.wikimedia.org/w/index.php?curid=69916624
2. By Hbf878 - Own work, CC0, https://commons.wikimedia.org/w/index.php?curid=65755645

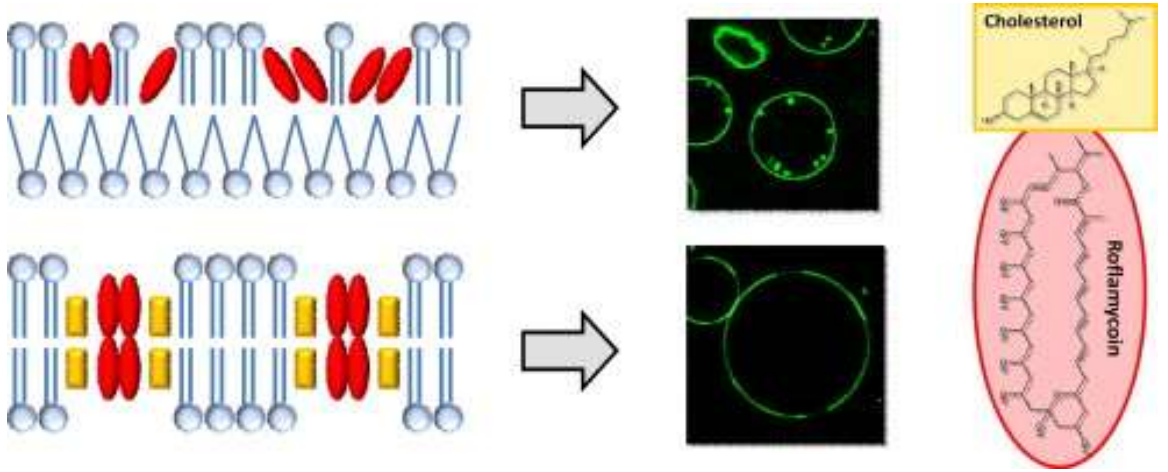
Natural products



Roflamycoin

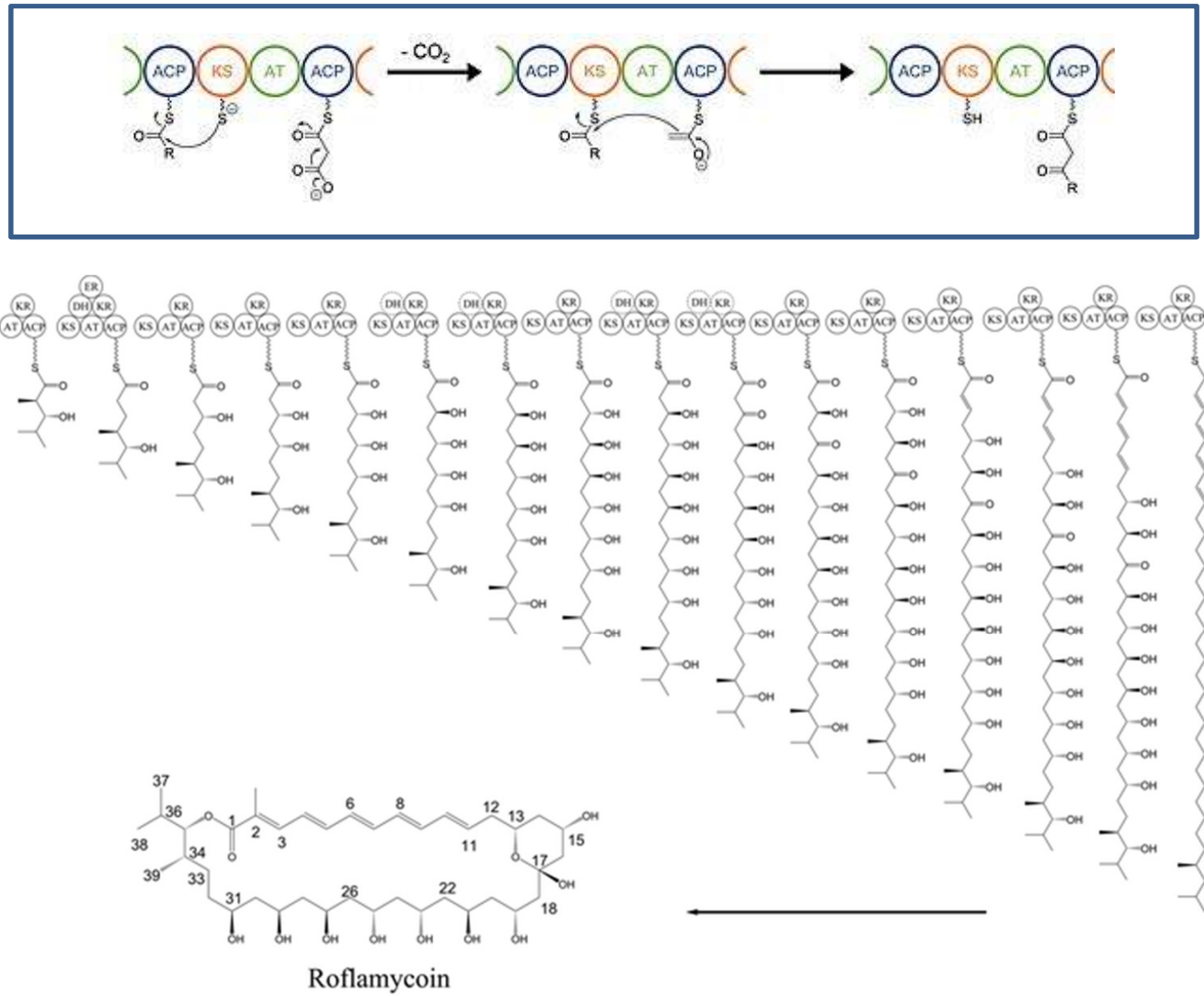
Roflamycoin was isolated from Streptomyces roseoflavusas an antifungal agent (1981). Roflamycoin is an unusual member of the polyene macrolide antibiotics as it is the only oxopolyene macrolide that has been shown to form well-defined ion channels





J. Am. Chem. Soc. **1997,** *119,* 2058-2059

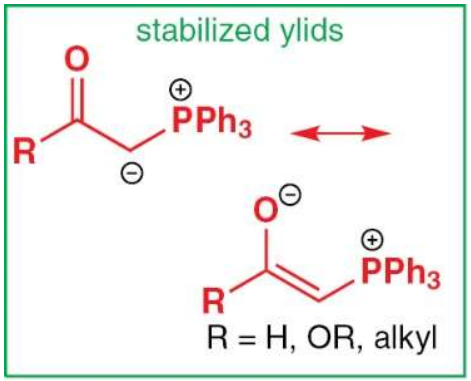
Proposed biosynthetic pathway of Roflamycoin

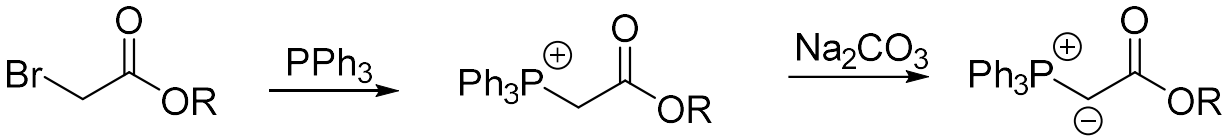


AT, acyl transferase; ACP, acyl carrier protein; KS, ketosynthase; DH, dehydratase; KR, ketoreductase; ER, enoyl reductase; TE, thioesterase.

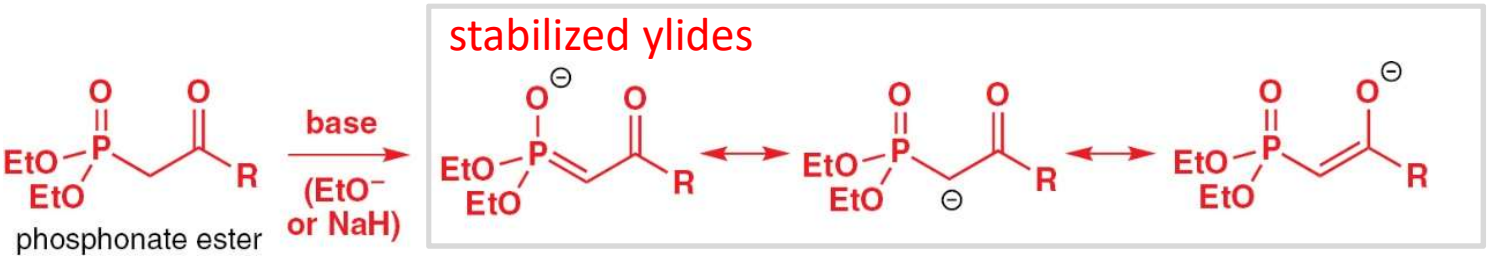
Ylides for Wittig reactions (from previous lecture)

Those with conjugating or anion stabilizing substituents adjacent to the negative charge (stabilized ylides)

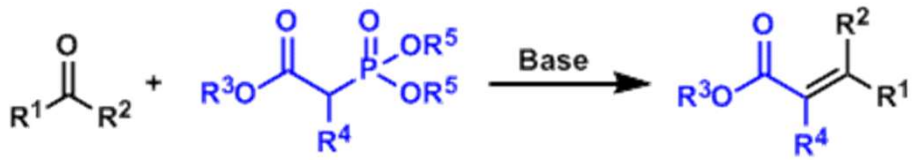




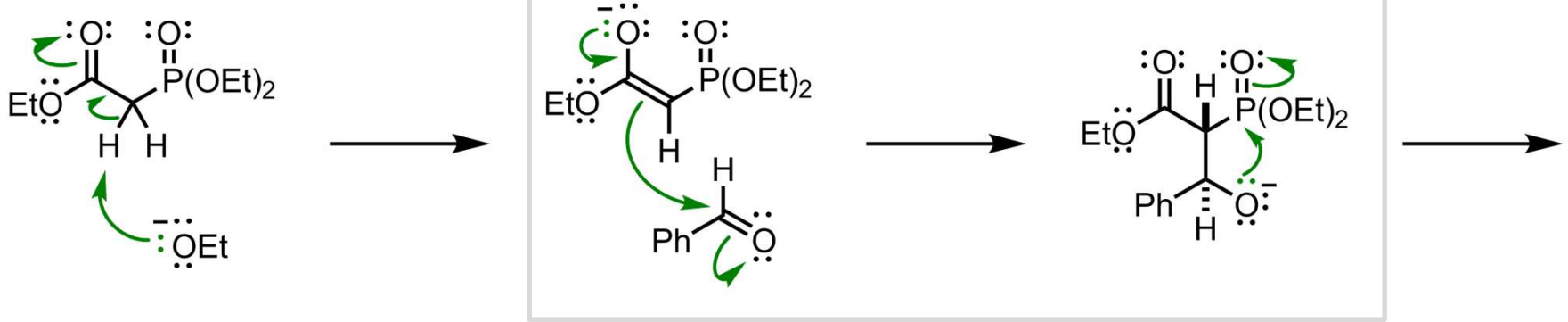
Horner-Wadsworth-Emmons (HWE) reactions with stabilized ylides.



Horner-Wadsworth-Emmons (HWE) Reaction (lecture 6)

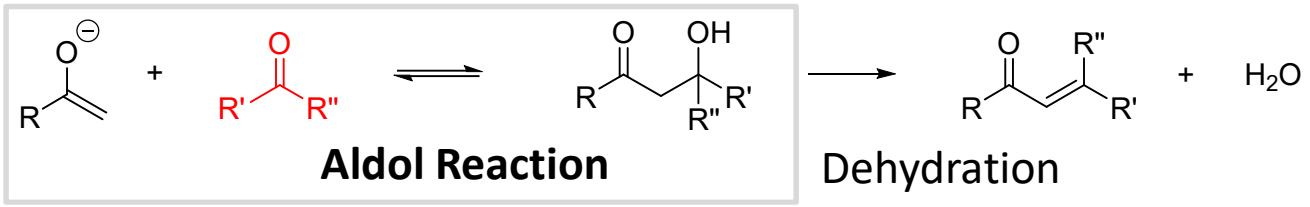


Mechanism



Aldol Reactions

Aldol reactions: Reactions between an enolate (or enol) and a carbonyl compound (aldehyde or ketone)



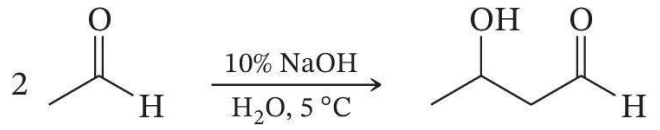
I. The aldol reactions
   - Under basic conditions
   - Under acidic conditions

II. Crossed-aldol reactions
   - Lithium enolates
   - Silyl enol ethers
   - Weak bases

III. Intramolecular aldol reactions

1.1 The aldol reactions under basic conditions

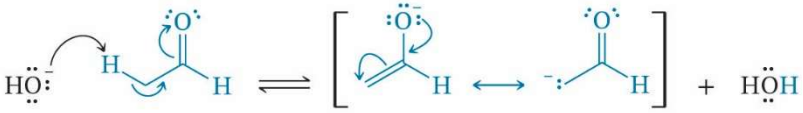
Treatment of acetaldehyde with a base, such as NaOEt or NaOH (catalyst) leads to rapid and reversible formation of 3-hydroxybutanal, known commonly as aldol (aldehyde+ alcohol), hence the general name for this reaction.



3-Hydroxybutanal
(50%)

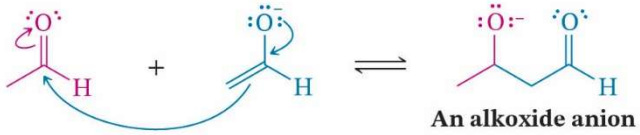
Self-aldol reaction (or dimerization)

Aldol mechanism (basic conditions)



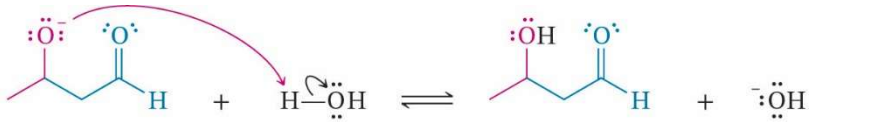
Enolate anion

In this step, the base (a hydroxide ion) removes a proton from the α carbon of one molecule of acetaldehyde to give a resonance-stabilized enolate.



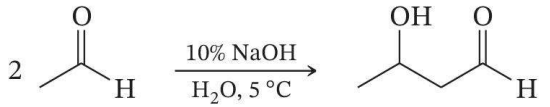
An alkoxide anion

The enolate then acts as a nucleophile and attacks the carbonyl carbon of a second molecule of acetaldehyde, producing an alkoxide anion.



Stronger base **The aldol product** **Weaker base**

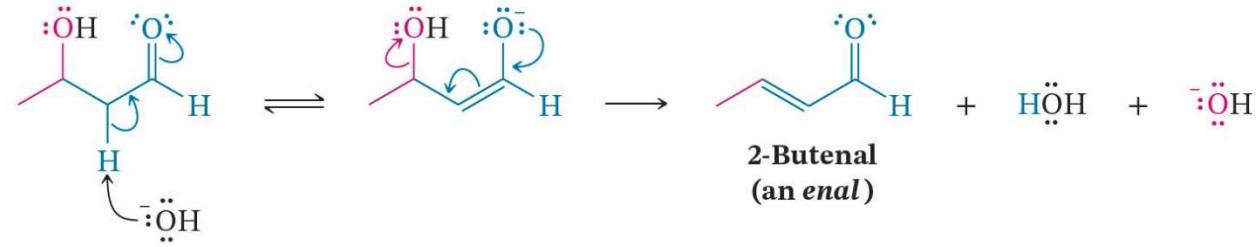
The alkoxide anion now removes a proton from a molecule of water to form the aldol product.



3-Hydroxybutanal (50%)

Dehydration (under basic condition)

E1cB (cB = conjugate base) via enolate intermediate



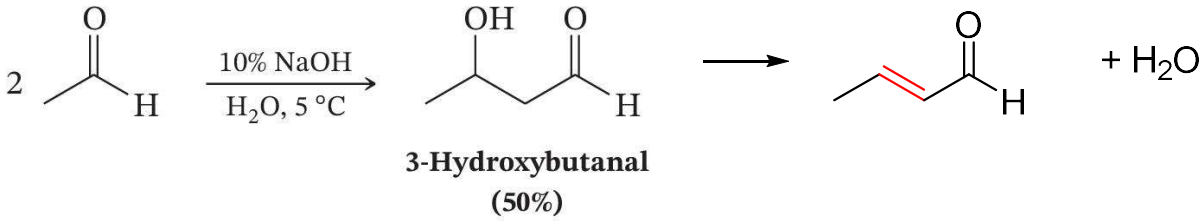
The α hydrogens are acidic.

The double bonds of the alkene and carbonyl groups are conjugated, stabilizing the product.

Dehydration of the aldol addition product leads to an α,β-unsaturated carbonyl compounds. The stability of the conjugated system drives the reaction forward

Aldol reactions: possible dehydration products

Based-catalyzed aldol reactions sometimes give the **aldol** and sometimes the **elimination product**. The more vigorous conditions (stronger based, higher temp, longer reaction time) is more likely to give the elimination product.

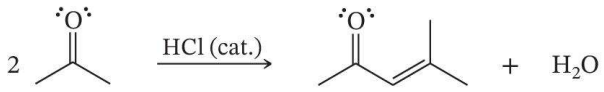


3-Hydroxybutanal (50%)

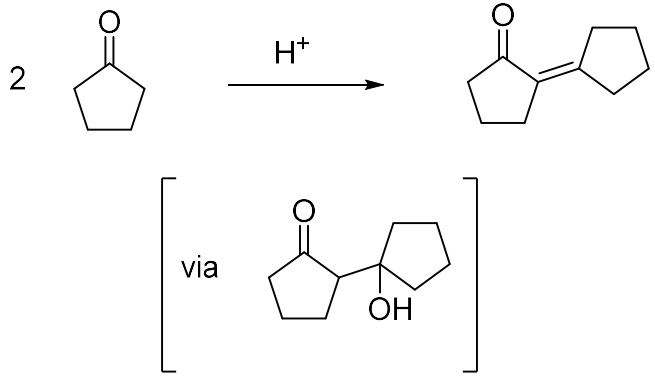
The term "condensation" refers to the combination of two molecules with a net loss of a small molecule (e.g. water, methanol etc). For example: Aldol condensation or Claisen condensation

1.2 Aldol reactions: acidic conditions

Aldol reactions can occur under acidic catalysis. The reaction generally leads to α,β-unsaturated carbonyl compound (**dehydration product**).

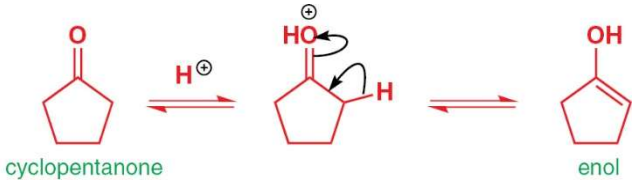


4-Methylpent-3-en-2-one

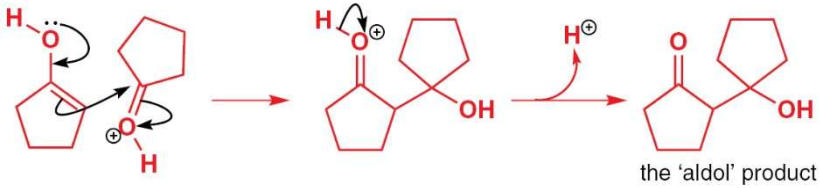


Mechanism: acidic conditions

the acid-catalysed enolization step

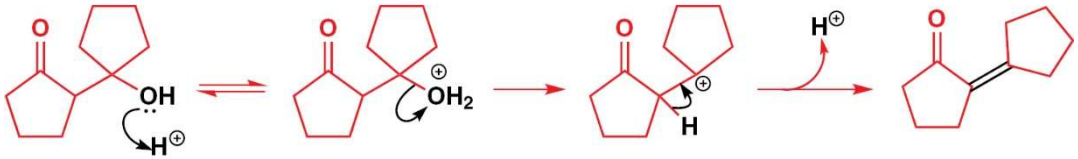


the acid-catalysed aldol addition step

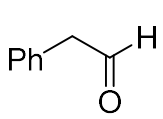


Dehydration (fast reaction, difficult to avoid under acidic conditions)

the acid-catalysed dehydration step (E1 elimination)

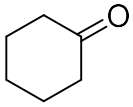


Aldol reactions



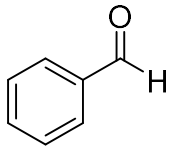
NaOH ⇌

(Aldol product)



NaOH ⇌

(Aldol product)



\+



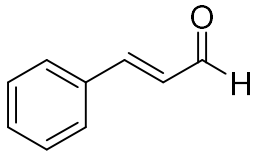
NaOH ⇌

(Dehydration product)

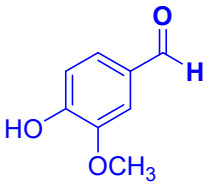
Natural almond oil

Second most important flavor after vanillin (vanilla main component): 7000 tons of synthetic and 100 tons of natural compound per year (2010).

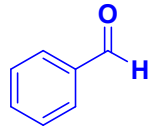
The 20 tons are isolated from the kernels of apricots, peaches, prunes and bitter almonds.
80% of what is called natural almond oil is produced by a chemical method, from a natural compound called cinnamaldehyde, which is cinnamon flavor. Cinnamaldehyde can be extracted in large quantities from cinnamon plant, which is commonly called cassia oil.



cinnamaldehyde



vallinin



benzaldehyde

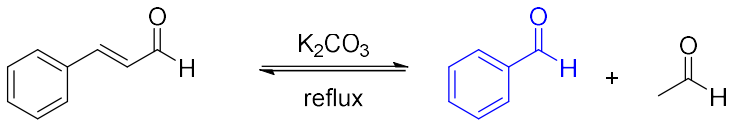




Innovation in food engineering : new techniques and products. Passos, Maria Laura., Ribeiro, Claudio P. Boca Raton, FL: CRC Press. 2010. p. 87.

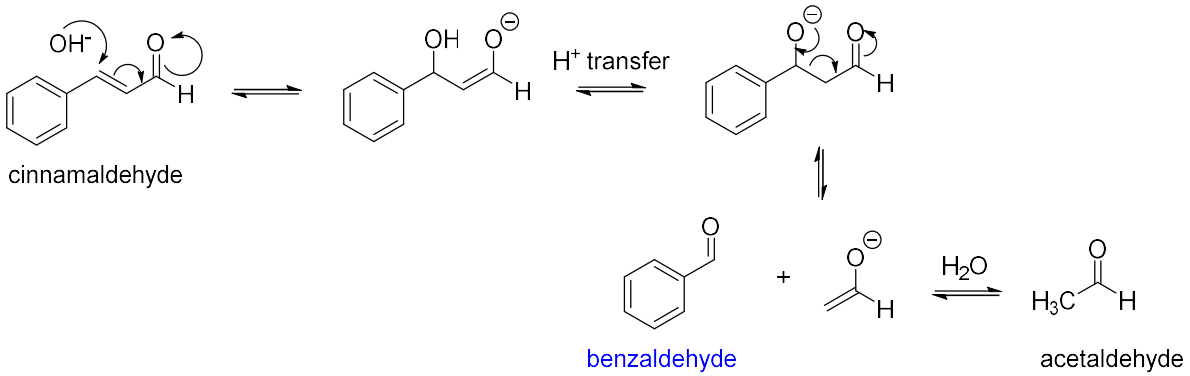
The aldol reaction is reversible: retro-aldol reaction

Most of the natural benzaldehyde is obtained by **retro-aldol** reaction of natural cinnamaldehyde.



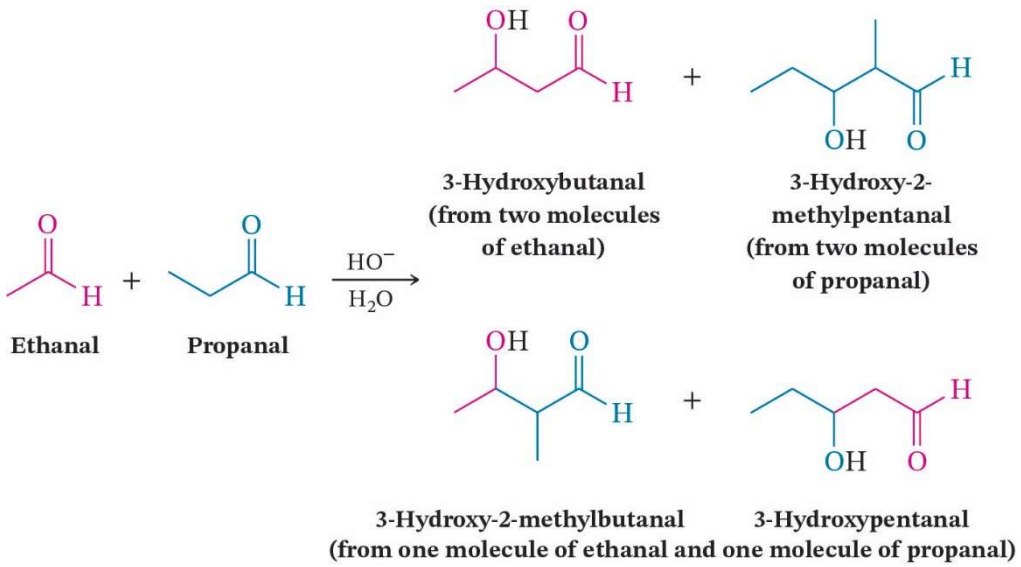
cinnamaldehyde benzaldehyde

Mechanism: retro-aldol reaction



cinnamaldehyde benzaldehyde acetaldehyde

2. Crossed-aldol reactions: problem



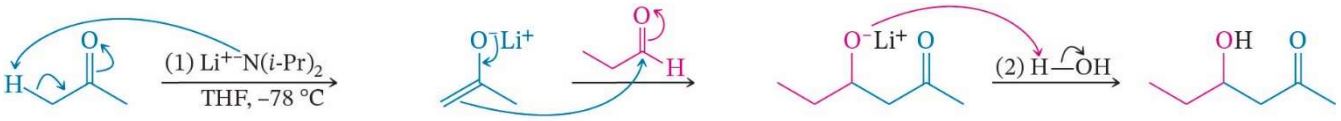
How to get 1 crossed aldol product selectively?

Crossed aldol condensations are aldol condensations between different carbonyl reactants

1. Lithium enolates (Pre-formation)
2. Silyl enol ethers
3. Weaker bases (Thermodynamic deprotonation)

2.1 Crossed-aldol reactions: with lithium enolate

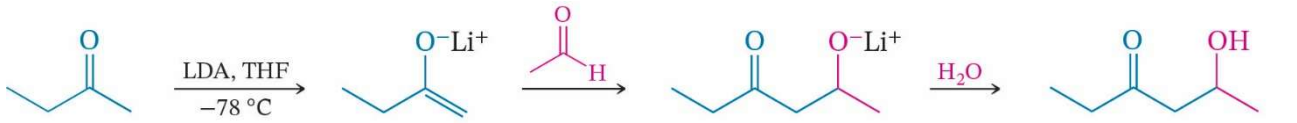
Directed aldol reaction: use LDA to form enolate from 1 partner (typically a ketone or an ester, aldehyde not likely to work), then add the second partner (an aldehyde or ketone).



The ketone is added to $Li^{+-}N(i\text{-}Pr)_2$ (LDA), a strong base, which removes an α hydrogen from the ketone to produce an enolate.

The aldehyde is added and the enolate reacts with the aldehyde at its carbonyl carbon.

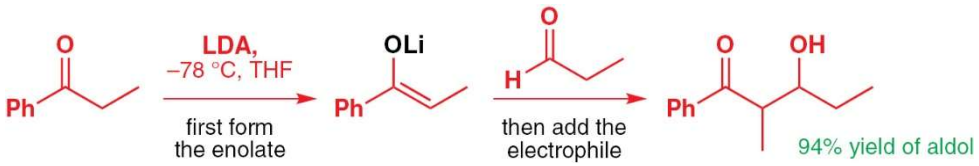
An acid–base reaction occurs when water is added at the end, protonating the lithium alkoxide.

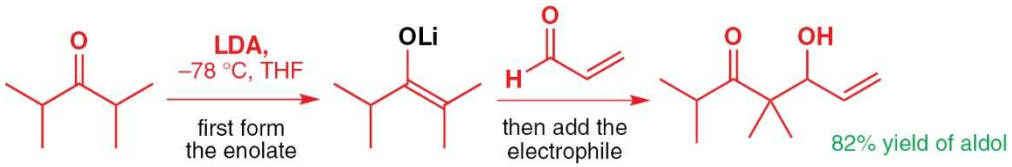


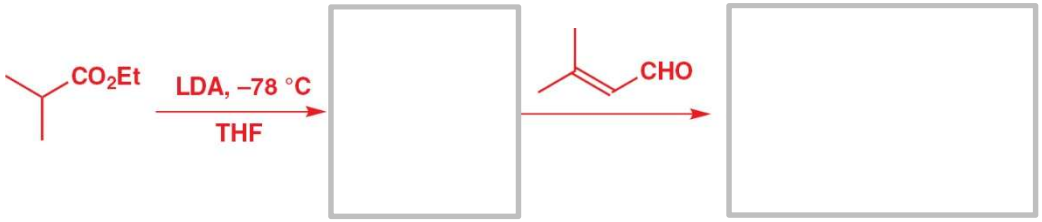
75%

A single crossed aldol product results.

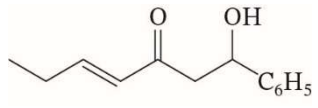
Examples

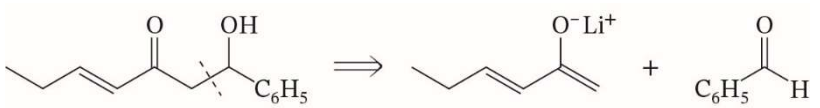

O
Ph
LDA,
–78 °C, THF
first form
the enolate
OLi
Ph
O
H
then add the
electrophile
O
OH
Ph
94% yield of aldol

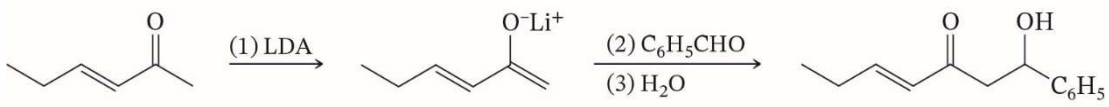

O
LDA,
–78 °C, THF
first form
the enolate
OLi
O
H
then add the
electrophile
O
OH
82% yield of aldol


CO2Et
LDA, –78 °C
THF
CHO

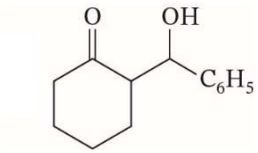
Aldol reaction with LDA

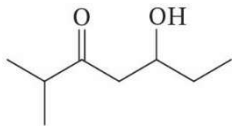






Practice



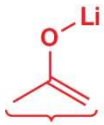


2.2 Silyl Enol Ethers (TMS-enolate)

Silicon is less electropositive than lithium, and silyl enol ethers are more stable, less reactive, than lithium enolates. Silyl enol ethers can be purified and characterized



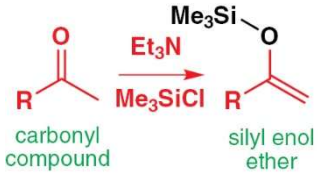
silyl enol ether

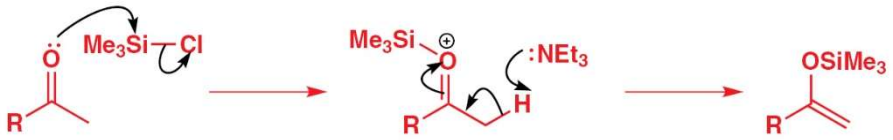


lithium enolate

Preparation

1. Carbonyl compounds + TMSCl, NEt_3



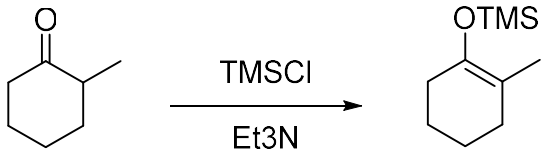


Bond Energy

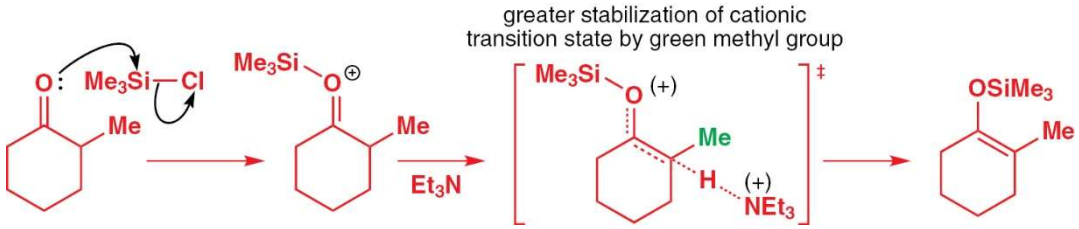
- **Si–O Bond:** 452kJ/mol
- **Si–C Bond:** 318kJ/mol

Preparation of silyl enol ethers

Carbonyl compounds + TMSCl, NEt_3: in the reaction with unsymmetrical ketones, the more substituted enol ether is formed (**thermodynamic TMS-enolate**)



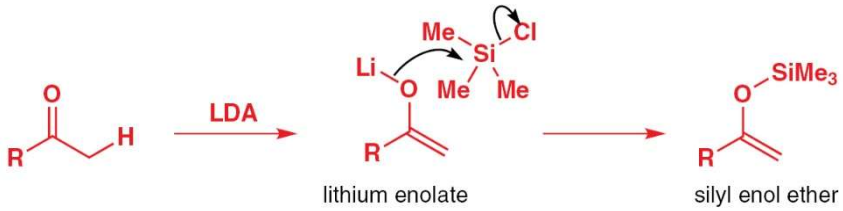
Explanation:

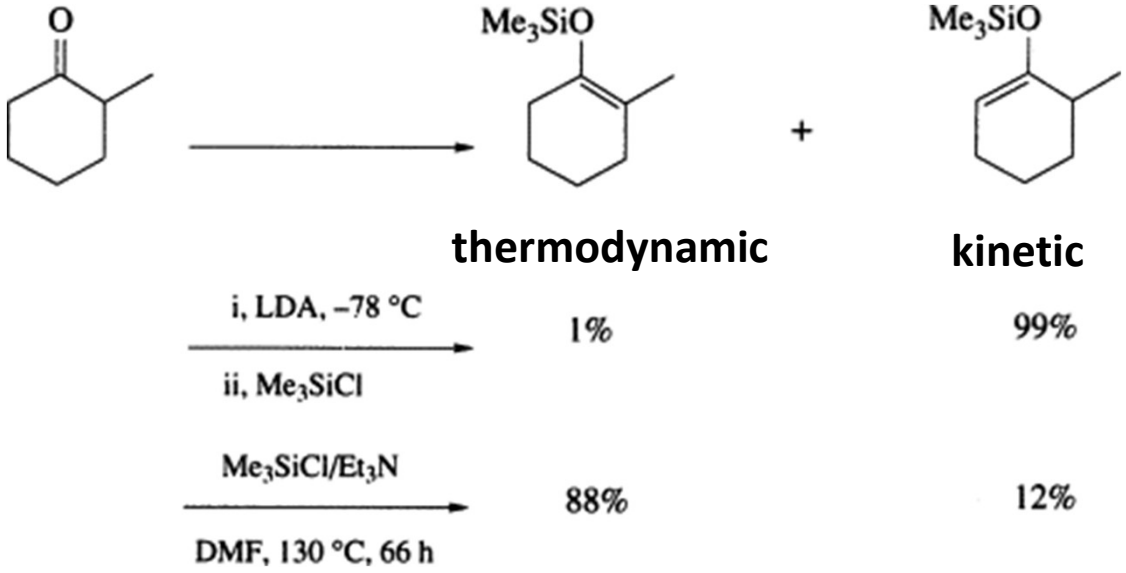


Silyl enol ethers vs lithium enolate

Preparation

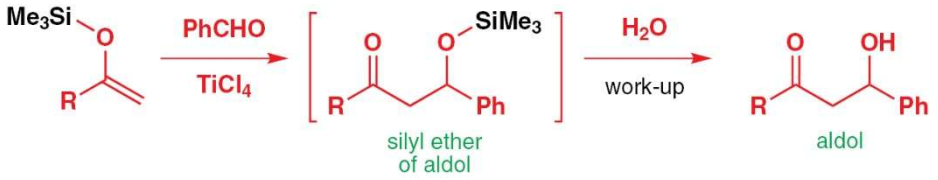
2. Lithium enolate + TMSCl → silyl enol ether (**kinetic TMS-enolate**)



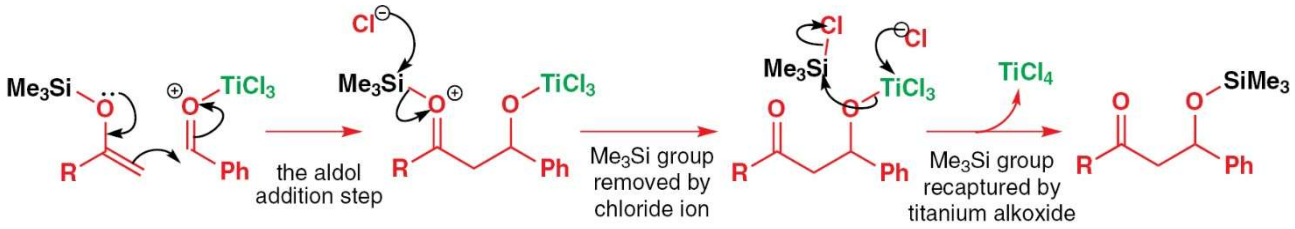


Mukaiyama aldol reaction

Silyl enol ethers react with aldehyde and ketones but require a Lewis acid catalyst, commonly Ti(IV).

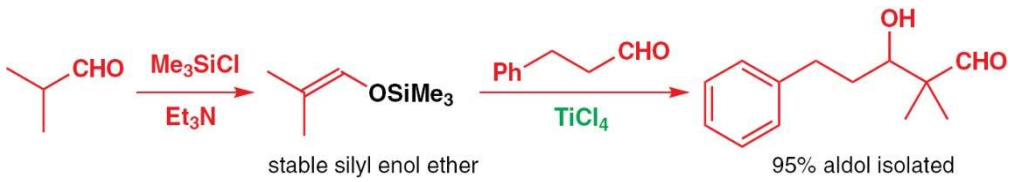


A possible mechanism:

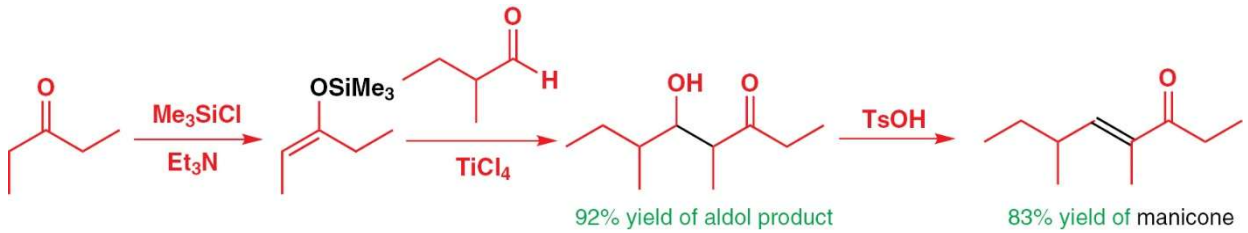


Mukaiyama aldol: examples

Aldehyde + aldehyde

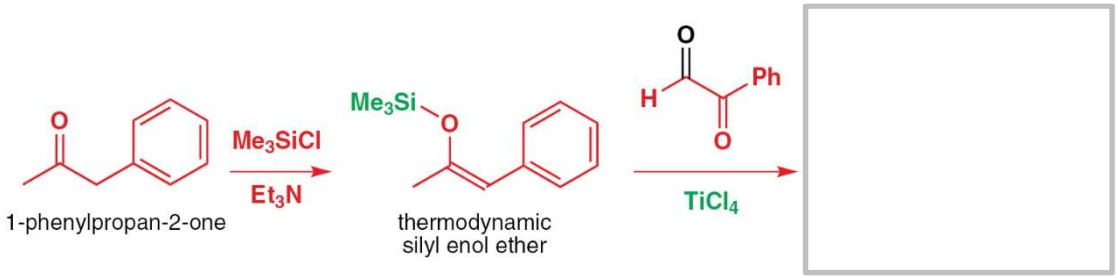


Ketone + aldehyde, follow by dehydration



Mukaiyama aldol: examples

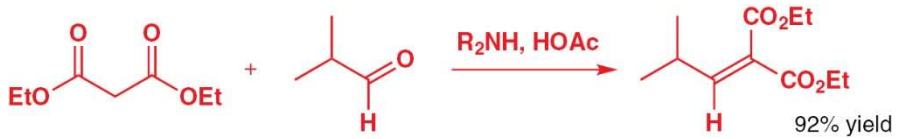
Ketone (thermodynamic enol ethers)



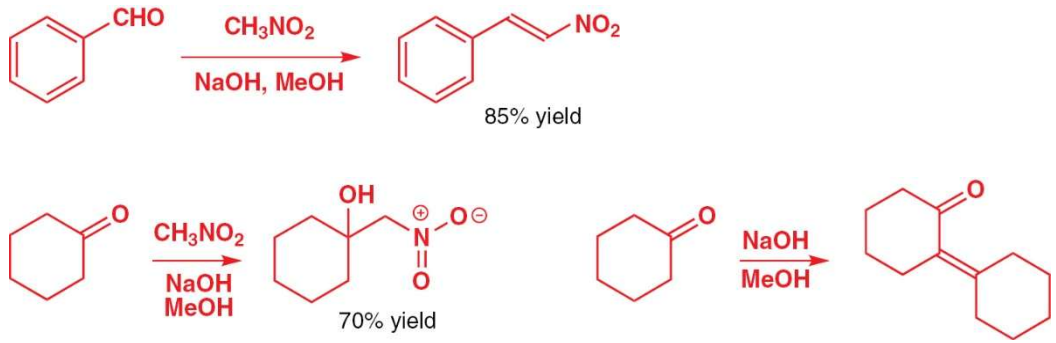
2.3 Crossed aldol reaction using “weak” bases

Case 1: One crossed aldol partner is much more acidic (Thermodynamic deprotonation)

a. 1,3-dicarbonyl compounds (more acidic than mono-carbonyl compounds)



b. Nitro compounds (more acidic than mono-carbonyl compounds)

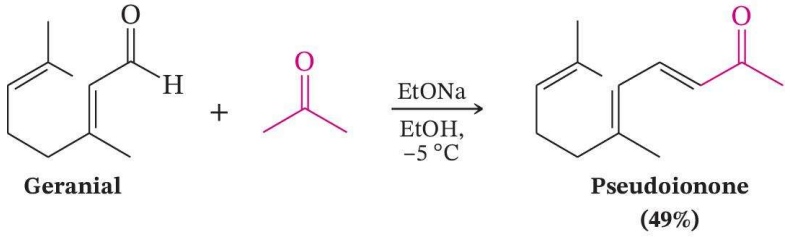


The aqueous pKa of nitromethane is approximately 10.2.

Crossed aldol reaction using “weak” bases

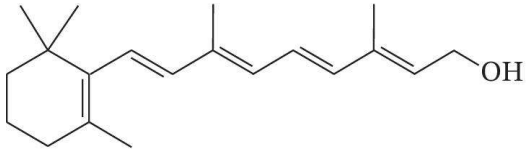
Case 2: One partner does not form enolate but is a better electrophile (e.g. aldehyde without alpha-H)

Commercial synthesis of vitamin A: crossed aldol reaction between geranial and acetone

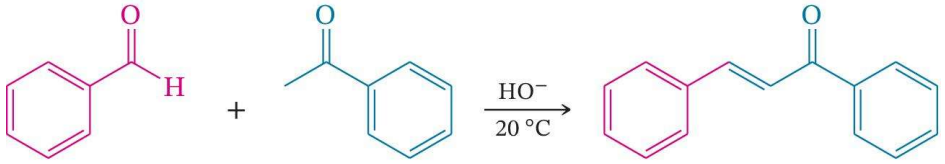


Geranial

Pseudoionone
(49%)



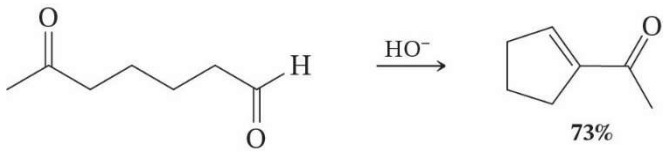
Vitamin A



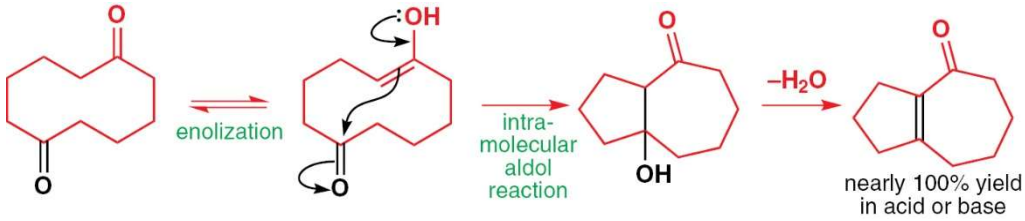
1,3-Diphenylprop-2-en-1-one
(benzalacetophenone)
(85%)

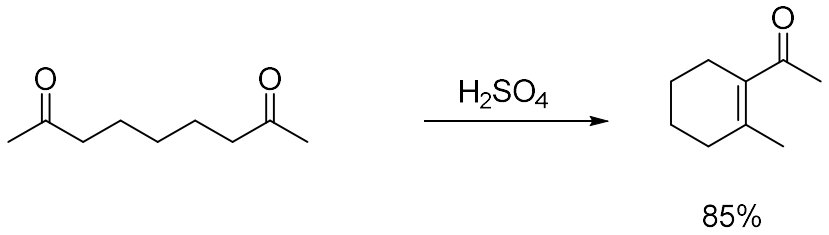
III. Intramolecular aldol reactions

The aldol reaction offers a convenient way to synthesize **5 or 6-membered rings** (and sometimes larger). The starting materials contain 2 carbonyl groups (dialdehyde, keto aldehyde or diketone)

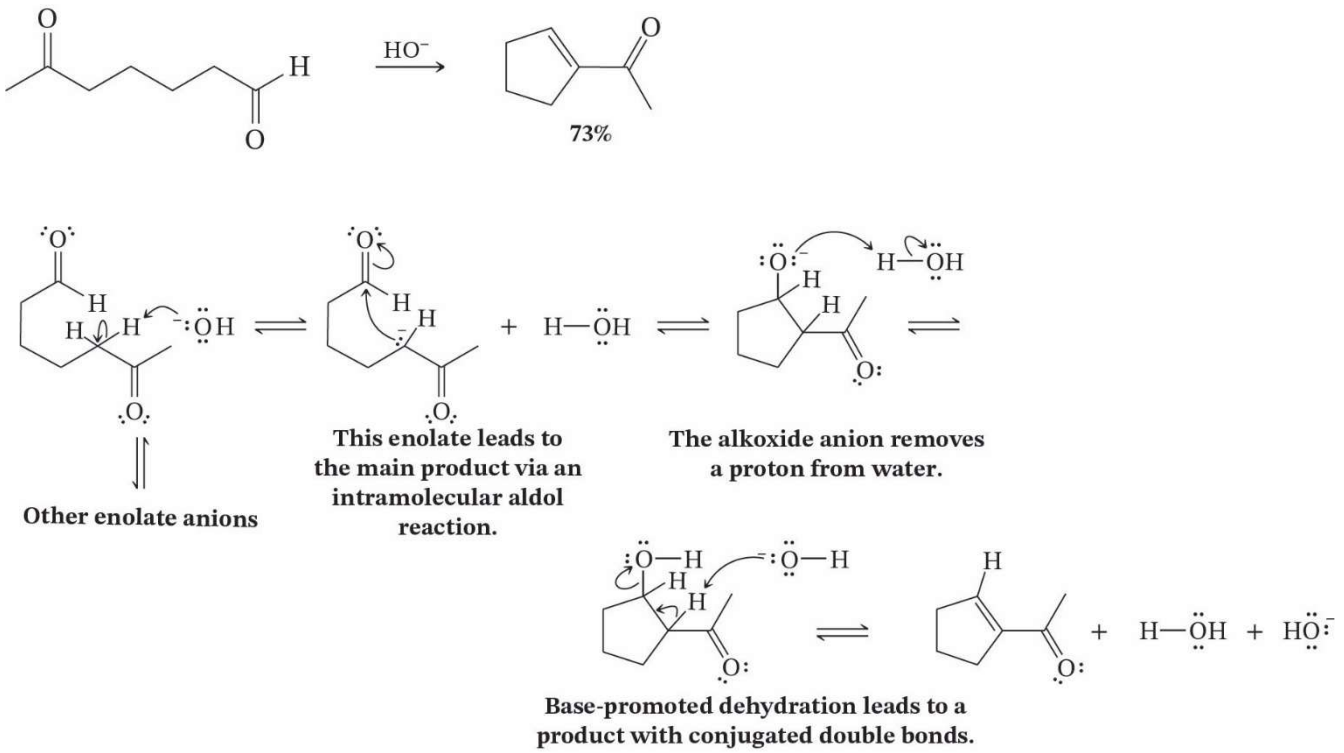


Aldehyde is the better electrophile





Mechanism: Intramolecular aldol reactions

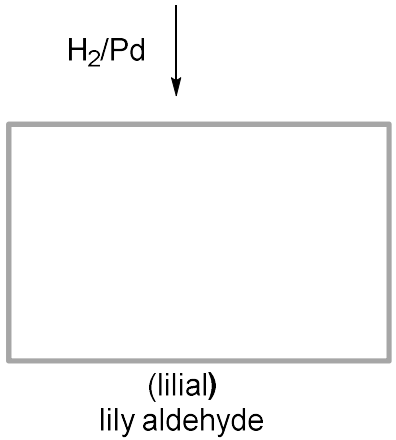


Synthesis of lily aldehyde





The (R)-enantiomer has a strong floral odor, reminiscent of cyclamen or lily of the valley; whereas the (S)-enantiomer possesses no strong odor

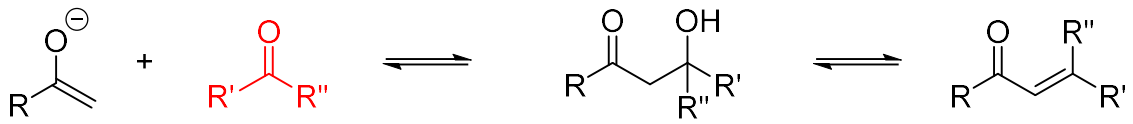


(lilial)
lily aldehyde

Photo by Ivar Leidus - Own work, CC BY-SA 4.0, https://commons.wikimedia.org/w/index.php?curid=118806658

IV. Reactions of enolates with various electrophiles

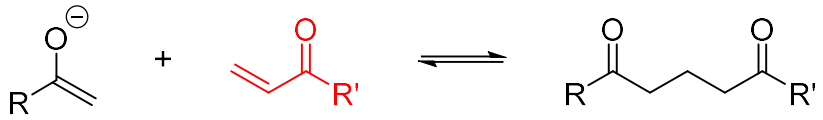
1. Aldol reactions: Enolate + aldehyde or ketone



2. Claisen reaction: Enolate + ester



3. Conjugate addition (**Michael addition**): Enolate + α,β-unsaturated carbonyl compounds



Summary

- Aldol reaction combines two carbonyl compounds (e.g. aldehydes or ketones) to form a new β-hydroxy carbonyl compound.
  - Under basic conditions
  - Under acidic conditions

- Crossed-aldol reactions
  - Lithium enolates
  - Silyl enol ethers
  - Weak bases

- Intramolecular aldol reactions
- Reactions of enolates with various electrophiles